Ball-Free Game State Reconstruction

Complete Easy Explanation, Real-Life Analogies & Viva Cheat Sheet (Slides 5 & 6)

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1. Project ka Asli Aim & Big Picture (Super Simple Language)

> ***Exact Aim (Official):*** *"To reconstruct football game state (ball possession and tactical advantage) solely from player tracking geometry without ball telemetry, and to systematically benchmark and mitigate broadcast tracking noise."*

Q: Hum ball tracking kyun nahi kar rahe?

TV broadcast video mein football bohot choti hoti hai. Match ke dauran ball aksar player ke pairon ke peeche chhip jaati hai, camera ke frame se bahar chali jaati hai, ya motion blur ki wajah se computer vision se detect nahi hoti. Isliye humne ball ka data intentionally ignore kar diya taaki hum bina ball ke bhi game analyze kar sakein.

Q: Yeh "Spatial Geometry" ka kya scene hai? (Real-Life Analogy):

Spatial Geometry ka matlab hai **"Ground par 22 players ke beech ki aapsi doori, unka spread aur banne wale shapes."**
Example: Maan lo match mein ball nahi dikh rahi, par aap dekhte ho ki ground ke beech mein ek **Blue jersey wala player aage daud raha hai**, uski taraf **2 Red defenders tezi se jhapattne aa rahe hain**, aur **1 Blue teammate support dene piche daud raha hai**. Aap bina ball dekhe bhi 100% bol sakte ho ki ball usi beech wale Blue player ke paas hai! Bas yahi calculation computer se mathematical shapes (**Voronoi & Delaunay graphs**) ke zariye karwane ko Spatial Geometry bolte hain.

2. Slide 5: Data, Schemas & Noise Engine (Challenges & Solutions)

Slide Point & Problem	Simple Explanation & Solution
1. Strict Anti-Leakage Boundary *Risk of model cheating*	Agar hamara model galti se bhi ball ka data ya match event logs padh leta, toh research fake ho jaati. **Solution:** Humne code mein sakht whitelist lagayi ki model sirf player coordinates (X, Y) padh sake. Event logs sirf aakhiri evaluation ke waqt use hote hain.
2. Multi-Source Tracking Discrepancies *Metrica vs SkillCorner*	Metrica 25 FPS par [0, 1] normalized scale mein tha, jabki SkillCorner 10 FPS par meters mein tha aur alag stadiums ki lambai (104m, 105m, 106m) alag thi. **Solution:** Stadium metadata se dynamic formula banaya jo dono ko ek unified Canonical Schema (v0.2.0) mein convert karta hai.
3. Degradation Compounding & Collapse *Noise balance issue*	Simulation mein random missing frames aur lambe gaps ko milane par 76% data gayab ho gaya aur trajectory toot gayi. **Solution:** Step 67 mein noise level recalibrate kiya (Severe gap prob 0.015 -> 0.010) taaki realistic monotonic ladder (Clean < Mild < Moderate < Severe) bane.
4. Identity Consistency & Tracklet Splits *Player ID confusion*	Players takrane par camera ID swap kar deta hai. Agar code mein galti ho toh ek frame mein do "Player 10" ban sakte the. **Solution:** Deterministic ID swapping and fragmentation logic banaya jisse 0 duplicate keys ki guarantee mili.

3. Slide 6: Mitigation, Tactical Inference & System Constraints

Slide Point & Problem	Simple Explanation & Solution
1. Jitter Suppression vs. Curvature *Noise removal vs real sprint*	Camera glitches se player kaanpta (vibrate) lagta hai. Par jab player sharp cut/turn leta hai, woh real sprint hota hai. Zyada smooth karne se real turn gayab ho jata hai. **Solution:** Savitzky-Golay polynomial filter (W=7, p=2) use kiya jo fake jitter ko 69% hata deta hai par physical curvature preserve karta hai.
2. Ball-Free Possession Disambiguation *Tight scuffles & crowd*	Penalty box mein jab 3-4 players jhund banakar khade hote hain toh bina ball ke guess karna mushkil hota hai. **Solution:** Voronoi shapes, pressing density, aur 5-frame persistence window lagaya jisse possession prediction frame-by-frame flicker na kare.
3. Post-Hoc Tactical Advantage (TAS) *Measuring tactics mathematically*	Football tactics ke liye koi sensor nahi hota jo ground truth bata sake. **Solution:** Tactical Advantage Score (TAS) formula banaya jo space control, defensive compactness aur passing lanes ko [0, 1] mein score karta hai aur noise par robustness check karta hai.
4. Experimental Grid Scalability *180 automated test runs*	4 match windows x 5 seeds x 3 noise levels x 3 conditions = 180 total tests. Itni saari files ka track rakhna tough tha. **Solution:** Automated audit scripts likhe jo 60 unique runspecs ko bina kisi human error ke calculate aur aggregate karte hain.
5. Broadcast Perception & GPU Constraints *Heavy GPU unavailability*	Asli 4K video par deep learning (SoccerNet-GSR) chalane ke liye massive GPU clusters chahiye the jo access mein blocked the. **Solution:** Humne heavy video model ke bajaye mathematically simulated synthetic noise framework use kiya jo CPU par hi fast chal gaya.

4. Examiner / Viva Rapid-Fire Cheat Sheet (30-Sec Answers)

Q1: What is the main aim of your capstone project?
Answer: *"Sir, our aim is to reconstruct ball possession and tactical advantage purely from 22-player tracking geometry without relying on the ball, and to provide an auditable noise-mitigation pipeline that handles real-world broadcast tracking flaws."*

Q2: How do you know who has the ball if you don't track the ball?
Answer: *"Through Spatial Geometry. When a player receives the ball, nearby opponents close in to press (high pressure), teammates move closer to provide passing options (support), and the defending backline compresses. We compute dynamic Voronoi cells and Delaunay graphs to detect these spatial signatures."*

Q3: Why didn't you use raw video or deep learning on GPU?
Answer: *"Because video tracking models like SoccerNet-GSR ultimately output X/Y coordinates. Our research focuses on the downstream tactical layer. By evaluating on clean Metrica data, broadcast SkillCorner tracking, and controlled synthetic noise, we proved downstream robustness scientifically without GPU bottlenecks."*

Q4: What is the single biggest numerical achievement of your pipeline?
Answer: *"Our Pipeline A (bounded interpolation + Savitzky-Golay polynomial smoothing) suppressed velocity jitter by up to **69.0%** under severe noise and recovered possession F1-score from **0.235 back to 0.413 (+17.8% recovery)**."*